

Gesture-Controlled Robot

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Life can be wonderful if everything around us can be controlled by simple gestures. Gesture recognition technology helps us to interact with machines naturally without any additional device. Gestures are interpreted via mathematical algorithms and corresponding actions

initiated. Although this technology is still in its infancy, applications are beginning to appear. Kinect is one such application.

Though initially invented for gaming, Kinect is being used for different purposes. Kinect is a motion-sensing and speech recognition device developed by Microsoft for Xbox 360 video game console. The main idea was to be able to use a gaming console without

any kind of controller.

This project uses Kinect technology to capture, process and interpret human gestures for controlling the motion of a robot.

Circuit and working

Fig. 2 shows the block diagram of gesture-controlled robot. It comprises a Kinect sensor interfaced with a computer using USB port and a simple robot connected to the computer through USB-to-serial converter.

Kinect sensor. Kinect sensor is packed with an array of sensors and specialised devices to pre-process the information received. The Kinect and the computer—running Windows or Linux—communicate through a single USB cable.

Main features of Kinect sensor include:

Gesture recognition. It can recognise gestures like hand movements, based on inputs from an RGB camera and depth sensor.

Speech recognition. It can recognise spoken words and convert them into text, although accuracy strictly de-



Fig. 1: Kinect sensor (Courtesy: Wikipedia)

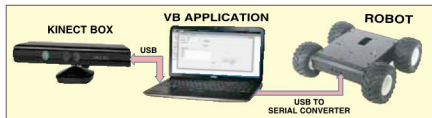


Fig. 2: Block diagram of gesture-controlled robot

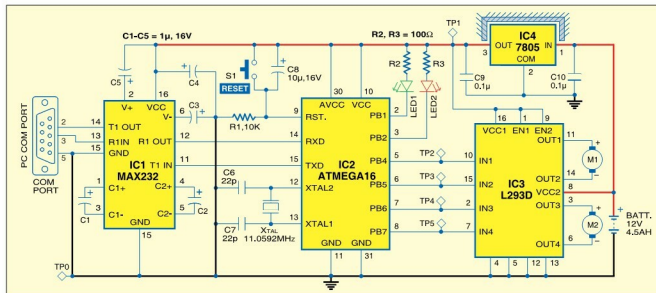


Fig. 3: Circuit of the robot

